

CALCULATION WORKSHEET
FOR CALCULATING THE AGRONOMIC RATE FOR THE LAND APPLICATION OF SEWAGE SLUDGE

Site No.: 2 Site Name: 21N9W20TC01 Site Location: 20-21N-9W

Procedure: A procedure used to calculate the agronomic rate for application of sewage sludge at which the nitrogen supplied by the sludge and available to the plant does not exceed the requirement for nitrogen of the crop or vegetation. To calculate the agronomic rate, the available ammonium nitrogen (NH_4 as N_{avail}), nitrate nitrogen (NO_3 as N_{avail}), organic nitrogen (Org as N_{avail}), must all be determined to calculate the total available nitrogen (TN_{avail}) in the sludge. The nitrogen needed (N_{needed}) by the crop is calculated basis the crop selected, expected yield, soil type, previous crop residual and nitrate nitrogen retained in the soil. Then the amount of nitrogen needed by the plant (N_{needed}) is divided by the total nitrogen available (TN_{avail}) to find the annual loading rate.

Step 1: From analysis of the sewage sludge to be land applied, determine the amount of each nitrogen compound, based on dry weight, in pounds per ton (lbs/ton)

Nitrogen Compound	Concentration of Nitrogen Compound (mg/kg)			Current Amount of Nitrogen in Sludge (lb/dry ton of sludge)
Total Kjeldahl Nitrogen (TKN as N)	<u>116,000</u> mg/kg	x 0.002	=	<u>232</u> lbs/ton TKN as N
Ammonium Nitrogen (NH_4 as N)	<u>84,600</u> mg/kg	x 0.002	=	<u>169.2</u> lbs/ton NH_4 as N
Nitrate Nitrogen (NO_3 as N)	<u>0</u> mg/kg	x 0.002	=	<u>0</u> lbs/ton NO_3 as N
Organic Nitrogen	TKN as N - NH_4 as N	x 0.002	=	<u>62.8</u> lbs/ton Organic N
Total Phosphorous (sludge) =	<u>33,900</u> mg/kg or ppm			

Step 2: Calculate the amount of ammonium-nitrogen available in the sewage sludge to be applied. Assume that the available fraction (K_v) is dependent upon operations at the site (see Table 1). Use the following equation:

$$\text{NH}_4 \text{ as } N_{\text{available}} = (\text{NH}_4 \text{ as N})(K_v)$$

Where,

$\text{NH}_4 \text{ as N}$ is the amount of ammonium nitrogen in the sewage sludge to the land applied, lb/ton

K_v is a volatilization factor for determining the availability of ammonium nitrogen based on how the sewage sludge is applied (see Table 1)

$$\text{NH}_4 \text{ as } N_{\text{available}} = \frac{169.2 \text{ lbs/ton}}{\text{from Step 1}} \times \frac{1}{K_v} = \underline{169.2 \text{ lbs/ton}}$$

Table 1. Factors for K_v

If Sewage Sludge is:

Factor K_v is:

Liquid and Surface Applied	0.25
Liquid and Incorporated into the Soil	1.00
Dewatered and Applied in any Manner	1.00

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Step 3: Calculate the amount of organic nitrogen available in the sewage sludge to be applied. The factor f , used for determining the amount of Org as N present due to mineralization, is provided below in Table 2. The value of F is dependent upon how the sludge is treated (i.e., aerobic digestion, composted, etc.).

Step 3A: Current Available Organic Nitrogen, Current Org as $N_{available}$. Current available organic nitrogen from this year's sludge is determined by the following equation:

$$\text{Current Org as } N_{available} = \text{Org as N (from Step 1)} \times F$$

Where,

Current Org as $N_{available}$ = the nitrogen which will be available this year from this year's sludge.
 Org as N = the organic nitrogen in the sewage sludge to be land applied, lb/ton.
 F = is the mineralization rate from Table 2

$$\text{Current Org as } N_{available} = \frac{62.8 \text{ lb/ton}}{\text{From Step 1}} \times 0.2 = 12.6 \text{ lb/ton}$$

Table 2. F Values				
Time After Sludge Application (Year)	Stabilized Primary and Waste Activated Sewage Sludge's, Fraction of Org as N	Aerobically Digested Sewage Sludge Fraction of Org as N	Anaerobically Digested Sewage Sludge Fraction of Org as N	Composted Sewage Sludge Fraction of Org as N
0 to 1	0.40	0.30	0.20	0.10

Step 4: Total available nitrogen in the sludge is then determined by adding together the resulting totals from Steps 2 and 3 to the amount of NO_3 as N in Step 1 (Assuming 100% of NO_3 as N is available). The result is the following equation:

$$\text{Total Nitrogen Available (TN}_{available}) = \text{NO}_3 \text{ as N} + \text{NH}_4 \text{ as } N_{available} + \text{Current Org as } N_{avail}$$

$$\text{TN}_{available} = \frac{0 \text{ lb/ton}}{\text{Step 1 / NO}_3 \text{ as N}} + \frac{169.2 \text{ lb/ton}}{\text{Step 2 / NH}_4 \text{ as } N_{avail}} + \frac{12.56 \text{ lb/ton}}{\text{Step 3 / Current Org as } N_{avail}}$$

$$\text{TN}_{available} = 181.76 \text{ lbs/ton of dry sludge. This is the total available nitrogen in the sewage sludge and is used as the denominator in the equation in Step 8.}$$

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Step 5: Nitrogen Credits. The available nitrogen credits from previous legume crops and existing residuals must be determined.

Step 5A: Previous Legume Crop, $N_{\text{previous crop}}$ If the crop for the previous year was a legume and was plowed under, there will be a previous crop nitrogen credit in the soil. Select the appropriate nitrogen credit based upon the data shown in Table 3.

$N_{\text{previous crop}} = 0 \text{ lbs/acre}$

Table 3. Nitrogen Credits from Legumes in Rotations	
Legume Crop	Nitrogen Credit
Alfalfa (1st year after)	
>80% stand	100-140 lbs/acre
60-80% stand	60-100 lbs/acre
<60% stand	0-60 lbs/acre
Alfalfa (2nd year after)	Half of 1st year credit
Sweet Clover	100-120 lbs/acre
Red Clover	40-80 lbs/acre
Soybeans*	30-60 lbs/acre
*(Allow 1 pound of N credit per bushel of yield. No credit for wheat double-cropped after soybean harvest.)	

Step 5B: Existing Nitrate Content of Soil, N_{residual} The nitrogen credit for the existing nitrate level in the soil can be accounted for by using the soil test nitrate results in the following equations. (See soil testing procedures for soil sampling methods.)

(Use the data for the top (nominal 6 inch) soil composite sample.)

Depth of Sample = 6 inches

$\text{NO}_3 \text{ as } N_{\text{soil}} = 0 \text{ mg/kg or (ppm)}$

$N_{\text{residual}} = \text{NO}_3 \text{ as } N_{\text{soil}} \times \text{Density of Soil}^*$

$N_{\text{residual}} = \text{NO}_3 \text{ as } N_{\text{soil}} \text{ mg/kg} \times (0.3 \text{ lb} \times \text{depth of sample, inches})$

$N_{\text{residual}} = \frac{0}{\text{NO}_3 \text{ as } N_{\text{soil}} \text{ (mg/kg)}} \times 0.30 \times \frac{6}{\text{Depth of Sample (inches)}}$

$N_{\text{residual}} = 0.00 \text{ Lbs / Acre}$

*300,000 Lb/acre-inch

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Step 5C: N/A Organic Nitrogen_{available}. Because the mineralization of organic nitrogen in sludge occurs over a long, time there will be a nitrogen credit for mineralization of previously applied sludge. This nitrogen credit is 0.5 of the previous year's calculated total available organic nitrogen.

a. $\frac{N/A}{PYCONA}$ lbs/ton. Step 3 on previous year's form.

b. $\frac{N/A}{PYSAR}$ dry tons / acre.

N/A Organic Nitrogen_{available} = (0.5) (PYCONA) (PYSLR)

c. $\frac{0}{PYONA}$ lbs / acre

Step 5D: Total Nitrogen Credits Summary

Total Nitrogen Credits = Previous Crop Credit + Existing Nitrate Content of Soil + Previous Organic Nitrogen

$$\text{Total } N_{\text{credits}} = \frac{0}{\substack{\text{Previous Crop} \\ \text{Credit (Step} \\ \text{5A)}}} + \frac{0}{\substack{\text{Existing Nitrate} \\ \text{Content of Soil} \\ \text{(Step 5 B)}}} + \frac{0}{\substack{\text{Previous} \\ \text{Organic} \\ \text{Nitrogen (Step} \\ \text{5C)}}} \text{ lbs/acre}$$

$$\text{Total } N_{\text{credits}} = \underline{0.00} \text{ lbs/acre}$$

Step 6: Crop Nitrogen requirement, N_{crop} . Determine the crop nitrogen requirement. The equation is:
From Table 4, select the crop to be grown and it's factor. From Table 5, select the soil factor.

Table 4.	
CROP	CROP FACTOR
Wheat	1.75 lbs N/bushel
Corn	1.35 lbs N/bushel
Oats	1.15 lbs N/bushel
Grain Sorghum/Milo	1.35 lbs N/bushel
Barley	1.50 lbs N/bushel
Soybeans	5.4 lbs N/bushel
Alfalfa	56 lbs N/ton
Orchard Grass	50 lbs N/ton
Brome Grass	33 lbs N/ton
Sunflowers	50 lbs N/1000 lbs seed
Tall Fescue	39 lbs N/ton
Forage Sorghum	9 lbs N/ton

Table 5. Soil Factor	
Type Soil	Factor
Sandy	1.1
All Other	1

Selected Crop is: corn

Crop Factor is: 1.35

Crop Yield : 250

Crop Nitrogen Requirement is:

$$N_{\text{crop}} = (\text{Crop Factor}) (\text{Yield}) (\text{Soil Factor})$$

$$N_{\text{crop}} = \frac{1.35}{\text{crop factor}} \times \frac{250}{\text{yield}} \times \frac{1}{\text{soil factor}} = \underline{337.5} \text{ lbs/acre}$$

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Step 7: Nitrogen Used, N_{used} . Based upon the previous calculations from Steps 5 and 6, the net amount of nitrogen used for the land application site can be calculated from these equations:

$$N_{used} = N_{crop} - \text{Total } N_{credits}$$

$$N_{used} = \frac{337.5}{\text{Step 6 } (N_{crop})} - \frac{0.00}{\text{Step 5D } (Total \text{ } N_{credits})} = \underline{337.50} \text{ lbs/acre}$$

Step 8: Determine the Agronomic Loading Rate (AGLR) for the sewage sludge. This is determined by dividing the nitrogen used by the plants (N_{used}) (from Step 7) by the total nitrogen available (TN_{avail}) (from Step 4) in the following equation:

$$AGLR \text{ (ton/acre)} = \frac{\text{Nitrogen needed by crops or vegetation } (N_{used}), \text{ lbs / acre}}{\text{Total Nitrogen Available } (TN_{avail}), \text{ lbs/ton}}$$

$$\frac{1.86}{AGLR} \text{ tons / acre}$$

Step 9: Determine the Approved Loading Rate (APLR) and the status of your agronomic rate for the land application of sewage sludge for the 2018 report.

$$APLR = (1.2) (AGLR)$$

$$\frac{2.23}{APLR} \text{ tons of dry sludge / acre}$$

$$\underline{2.22} \text{ tons of dry sludge applied / acre}$$

2018 agronomic rate for the land application summary.

Based on the information provided you have met the agronomic rate for the land application of sewage sludge for this reporting year. To ensure continued compliance with the 40 CFR Part 503 Domestic Sewage Sludge Re-use and Disposal Regulations, KDHE recommends that your facility continually monitors agronomic loading rate by calculating an estimated nitrogen loading rate for the next crop year for this site.

Step 10: From the soil analysis record the following parameters

0 - 6" depth sample

6 - 24" depth (profile) sample

Available

Phosphorous 47 mg/kg

Nitrate: 0 mg/kg

Exchangeable

Potassium 97 mg/kg

pH: 6.5 S.U.

Kansas State University, Cooperative Extension Service, Agronomy Program, currently recommends that pH be maintained between 6.5 and 7.0 for most field crops. Also, available phosphorus content of the soil should not be allowed to exceed 100 ppm to achieve maximum crop production. If high phosphorus contents are encountered, the disposal of sludge should be temporarily discontinued and an agronomist contacted to determine appropriate actions to be taken.

This completes your calculation of the Annual Nitrogen Loading rate for the Reporting Calendar Year. Please print a copy of this form as completed and submit it with your 503 sludge report. A copy of this report must also be maintained on file at your facility.